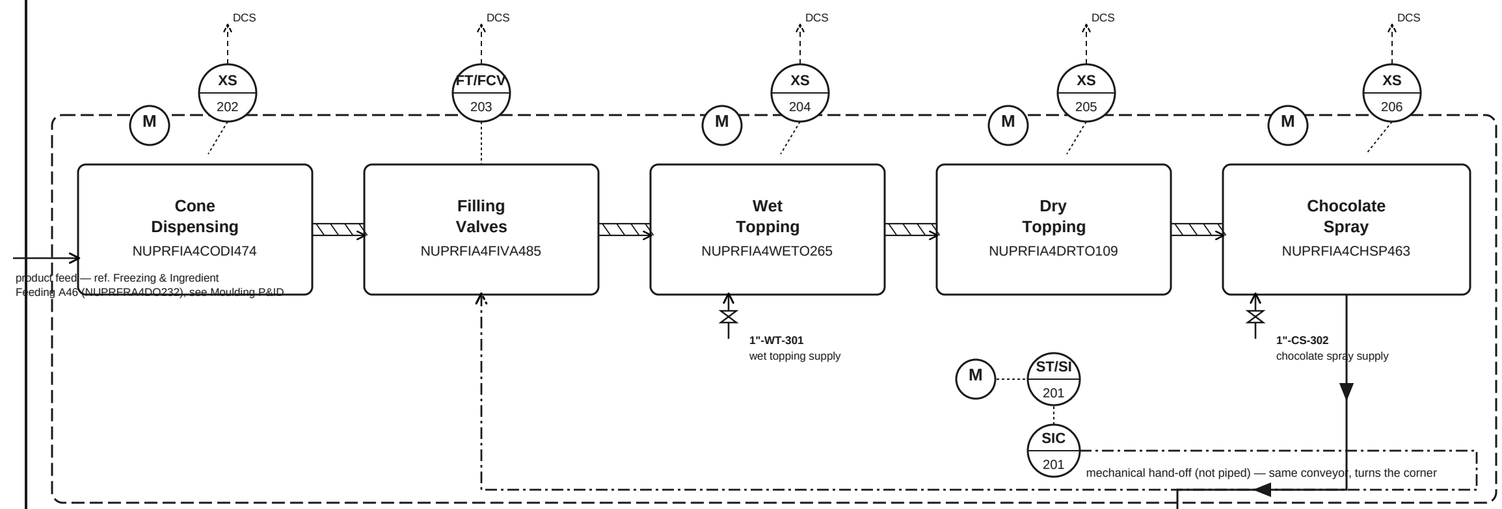
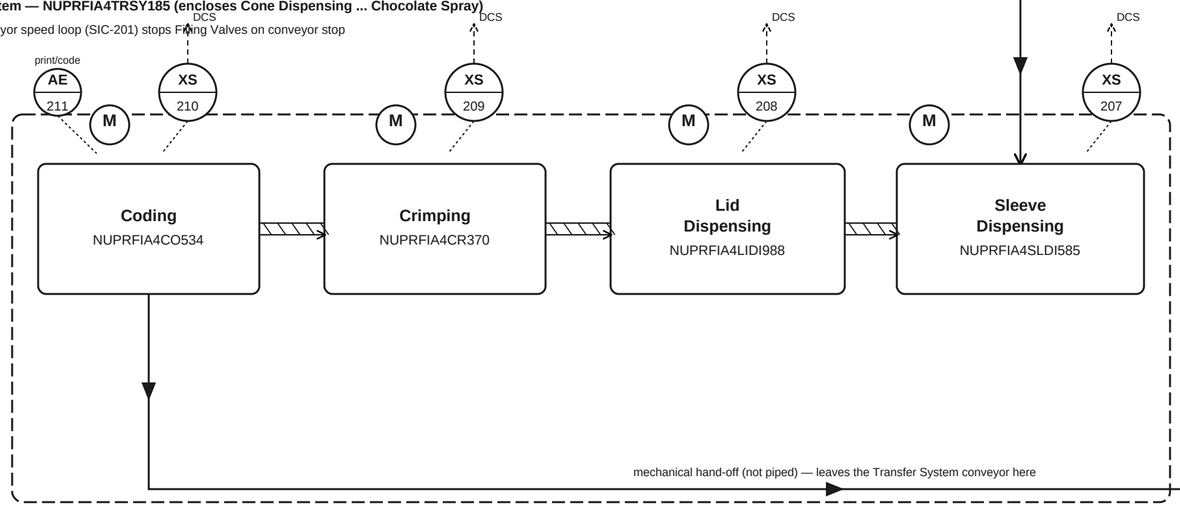
P&ID (PRELIMINARY) — NUREMBERG SITE — FILLING LINE A46 — ICE CREAM FACTORY DATA

Cognite Data Fusion Core Data Model | ISA-5.1 instrument tagging | For L100 practitioner training use only — not for construction

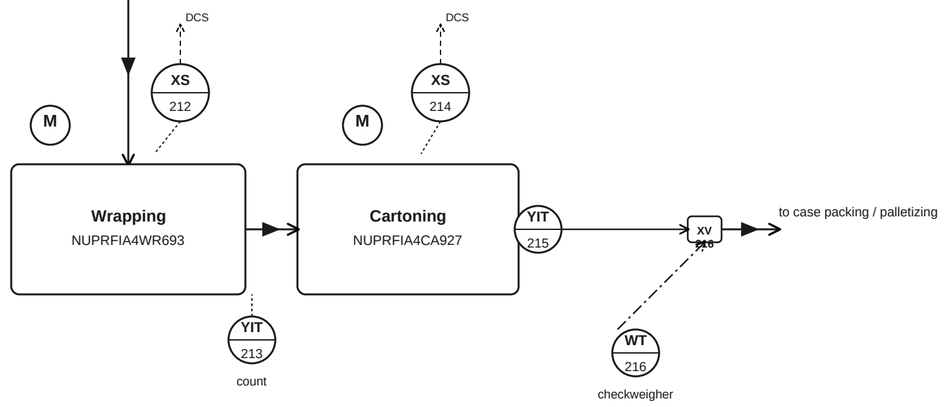


Transfer System — NUPRFIA4TRSY185 (encloses Cone Dispensing ... Chocolate Spray)

interlock: conveyor speed loop (SIC-201) stops Filling Valves on conveyor stop



Transfer System — NUPRFIA4TRSY185 (continued — encloses Sleeve Dispensing ... Coding)



same Cartoning model (SA-380CDF) as Moulding Line

Legend

Solid line + ►	Process piping / product flow (arrowheads = assumed direction)
Bowtie (on a line)	Manual isolation valve
size"-CODE-###	Line number: size-service code-sequence (WT=wet topping, CS=chocolate spray)
Short dashed line	Instrument signal to PLC / DCS
Dash-dot line	Safety interlock / control logic
Long dashed box	Equipment grouping enclosure (e.g. Transfer System conveyor)
Circle (M)	Electric motor / servo drive
FT / FCV	Flow transmitter / flow control valve (dosing, throughput)
AE	Analytical/vision inspection element (print & date-code check)
ST / SI / SIC	Speed transmitter / indicator / speed indicating controller
XS / YI	Motor state switch / status indicator
YIT / YC	Event or pulse counter (count variable)
WT	Checkweigher (quality variable)
XV / YV	Solenoid reject valve / pneumatic pusher

Notes / open items before this drawing is construction-ready

Line numbers and valve locations are illustrative — sizes, services, and sequence numbers are assumed, not taken from a line list or spec.

Process order (Cone Dispensing -> ... -> Cartoning) is inferred from standard cone/cup filling-line practice — the source data gives no explicit sequence, only a flat unit list.

Cone Dispensing -> ... -> Coding connectors carry NO line numbers or valves on purpose: that path is mechanical product transfer on the Transfer System conveyor (NUPRFIA4TRSY185), not piped flow.

Wet topping (WT-301) and chocolate spray (CS-302) supplies are inferred utility tie-ins, not present in the source data.

Filling Valves' product feed references the Freezing & Ingredient Feeding A46 cell rather than redrawing it — the Aging Tank / Continuous Freezer / Doser are a separate ISA-95 production line shared with Moulding Line A46 (see that P&ID for detail).

planned_status (Core Data Model time series) is a software/ planning variable only — intentionally not drawn on this P&ID.

This line has no vessels of its own — unlike Moulding Line A46, there is nothing here to vent, drain, or sample.

Wrapping and Cartoning here are the same unit types/specs as on Moulding Line A46 (Cartoning: model SA-380CDF, both lines).

Instrument loop numbers (201-216) are illustrative for this training exercise, not an assigned plant numbering series.

Source: Ice Cream Factory data, Nuremberg site, Filling Line A46, modeled via the Core Data Model (ProductionLine / Unit / Equipment).

Drawing No.	NUR-P&ID-103
Client / project	Ice Cream Factory — L100 practitioner training
Site	Nuremberg
Unit	Filling Line A46
Drawing type	P&ID — preliminary (ISA-5.1)
Data source	Core Data Model (ProductionLine, Unit, Equipment)
Status	For training use — not for construction
Rev.	0 — draft